

Using Stage-Discharge Data

Hydraflow Hydrographs Extension allows you to specify:

- Up to four unique culvert/orifice structures (including a perforated riser).

The culvert/orifice structures are labeled as A, B, C, and Prf Riser and can act like culverts or orifices depending on the inputs. See [Perforated Risers](#).

- Four unique weir structures.

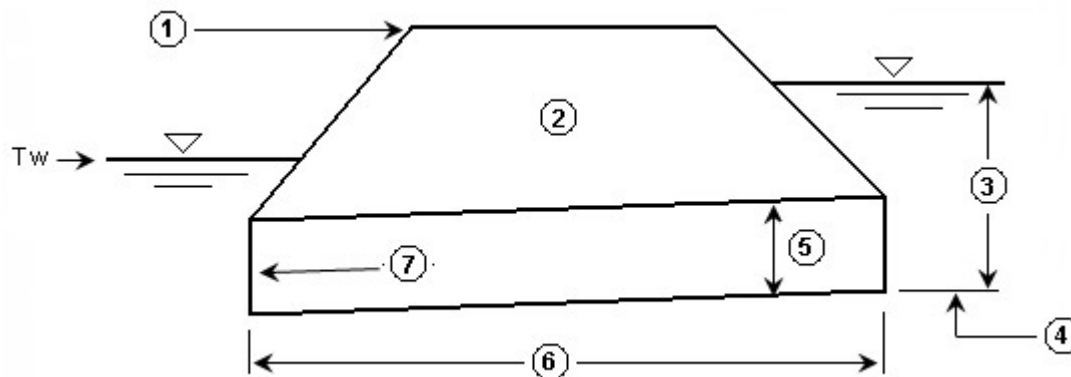
The four weir structures are referred to as weir A, B, C, and D. See [Weir Structures](#).

- An exfiltration component. See [Exfiltration](#).
- Known flow rates (Q_s).
- A single tailwater elevation. See [Tailwater Elevation](#).

Each structure can be used as an independent outflow structure or can be used in series or a multi-stage structure.

In computing the outflows, Hydraflow Hydrographs Extension treats the operation of these outlet structures as a function of stage or water surface elevation. Partial and full flow conditions are computed as well as inlet and outlet control.

The following illustration shows a profile of typical culvert.



Tw - Tailwater elevation	(3) Stage	(6) Pipe length
(1) Top of pond	(4) Invert elevation	(7) Flow
(2) Embankment	(5) Rise	

Best Practice

During the calculation process, both inlet control and outlet control are evaluated. Under inlet control, the discharge depends on the barrel shape, cross-sectional area, and inlet edge. The

inlet of the culvert controls the amount of flow that the culvert can handle. Under outlet control, the discharge depends on the slope, length and roughness of the barrel. Outlet control means that flow can enter the structure at a faster rate than it can exit.

Hydraflow Hydrographs Extension computes the discharge at each stage as established by the stage-storage table, including intermediate stage points that it generates, using both inlet and outlet control equation parameters. The smallest value is used as the discharge at that elevation. This is reflected on the table report as **ic** (inlet control) and **oc** (outlet control).

NOTE:

Hydraflow Hydrographs Extension does not assume full flow when the depth is partial.

Topics in this section

- [Perforated Risers](#)